IRWA Final Project 2026

Part 1: Text Processing and Exploratory Data Analysis

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Repository: https://github.com/bielmongeortiz/irwa-final-project
Code: `project_progress/part_1/IRWA_part_1.ipynb` and `myapp/search/preprocessing.py`

Part 1: Data preparation

The corpus contains 28,080 fashion products from Flipkart, each with a unique `pid`. The goal of this part is to turn the raw documents into a clean representation that can be indexed and ranked. The same function, `build_terms(text)`, is applied to documents at indexing time and to queries at search time, so both end up in the same term space.

1.1 Text pre-processing

Title, description and the other text fields go through seven steps, in this order:

#	Step	Why
1	Unicode normalization (NFKC)	Around 7k titles contain non-breaking spaces (`\xa0`) that break whitespace splitting.
2	Lowercasing	*Cotton* and *cotton* must be the same term.
3	Domain normalizations	Spelling variants become one term: `t-shirt`/`t shirt` → `tshirt`
4	Punctuation removal	Replaced by spaces.
5	Tokenization	Whitespace split
6	Stop-word removal	NLTK English list.
7	Stemming	Porter stemmer merges inflected forms

The table below shows the effect of each step on title + description:

Step	Tokens	Vocabulary
Raw (whitespace split)	1,017,425	13,521
lowercase, normalization, punctuation, tokenization	986,146	6,325
stop-word removal	687,984	6,205
stemming	687,984	4,741

Normalization removes more than half of the vocabulary. Stop-word removal mostly reduces the number of tokens, while stemming mostly reduces the vocabulary.

1.2 Output fields and processed representation

`preprocess_corpus` returns one row per product, saved in `data/processed_corpus.pkl`. Each row keeps:

- the 14 output fields required by the search engine, untouched, so results can always be shown with the original information
- the processed text fields, kept separately: `title_terms`, `description_terms`, `details_terms`, `brand_terms`, `sub_category_terms`

- `terms`, the concatenation of all of them, used to build the conjunctive (AND) inverted index
- the typed numerical attributes (prices, discount, rating, `out_of_stock`, `has_rating`)

1.3 category, sub_category, brand, product_details and seller

We first measured how distinctive each field is. A field whose value is shared by most documents has a near-zero IDF and cannot discriminate between products.

Field	Non-empty	Distinct values	Most frequent value
category	100%	4	Clothing and Accessories (96.6%)
sub_category	100%	24	Topwear (54.5%)
brand	92.8%	324	ECKO Unl (3.6%)
seller	94.1%	498	RetailNet (5.4%)
product_details (selected values)	98.1%	10,359	0.9%

The validation queries confirm that the title alone is not enough. For query 1 (*women full sleeve sweatshirt cotton*), *cotton* appears in none of the titles of the labelled products, but in all their descriptions and in 90% of their details. Over the whole corpus, an AND query returns 0 / 241 / 549 products when indexing *title + description + details*. Since approximately 40% of the products have no description, the details are often the only way to match them.

Decision:

- `category`: almost constant, so it is not indexed, it is kept as an attribute for filtering.
- `sub_category`: thick but meaningful, so it is indexed.
- `product_details`: only descriptive keys, and only their values. Style codes, care instructions, sizes or country of origin would add noise. Yes/No keys add the key name when the value is *Yes*.
- `brand`: indexed as its own field and not stemmed, because they are proper nouns. Many values are truncated in the dataset, which limits its usefulness.
- `seller`: not indexed. Users do not search by seller.

Merged field or separate fields? We keep both:

Approach	Pros	Cons
One merged field (`terms`)	Simple index, an AND query can match terms spread across fields.	Loses where a term appears, long descriptions dominate term frequencies.
Separate fields	Field weighting, filtering and faceting.	Larger index and more complex scoring, an AND query per field would be too strict.

1.4 out_of_stock, selling_price, discount, actual_price and average_rating

These fields are not indexed as textual terms, a number as a token only matches the exact same string, has no notion of order or range, and contaminates the vocabulary. They are parsed into typed values instead:

- prices *"2,999"* → `2999.0`, discount *"69% off"* → `69.0`, `out_of_stock` → boolean

- missing ratings (2,261) stay as NaN with a `has_rating` flag: no rating is not a bad rating, so we do not impute 0
- a missing discount is derived from the two prices. The reported and derived discounts differ by at most 1 point in 96.9% of products, so the reported value is trusted.

1.5 Assumptions and limitations

- The NLTK stop-word list includes negations, so a query like *not cotton* becomes *cotton*. We kept the standard list for simplicity.
- Porter produces non-words, which is harmless because queries are stemmed in the same way.
- Label noise**:** for query 2, 4 of the 10 relevant products say *Women* and never contain *men*, so no AND query can retrieve them.
- The domain normalizations were designed from the validation queries, other queries may need more.

Part 2: Exploratory data analysis (EDA)

The EDA describes the corpus and checks the assumptions behind the pre-processing. Each analysis ends with what it means for the search engine.

2.1 Corpus overview and missing values

About 96.6% of the products are *Clothing and Accessories* (the rest are 920 Footwear, 41 Bags, Wallets & Belts and 1 Toy). *Topwear* alone covers 54.5% (15,310 products), *Bottomwear* (3,666) and *Winter Wear* (2,493). Percentage of missing values per field:

description	average_rating	brand	seller	discount	actual_price	images
39.7%	8.1%	7.2%	5.9%	3.0%	2.8%	2.7%

Four in ten products have no description, so the title and the product details carry most of the searchable text.

2.2 Text length and vocabulary

Titles are short and regular (6.5 words on average). Non-empty descriptions have 49.4 words, in sentences of 15.7 words on average. The raw title + description has 13,521 different word forms; after pre-processing, all indexed fields together contain 5,053 distinct terms.

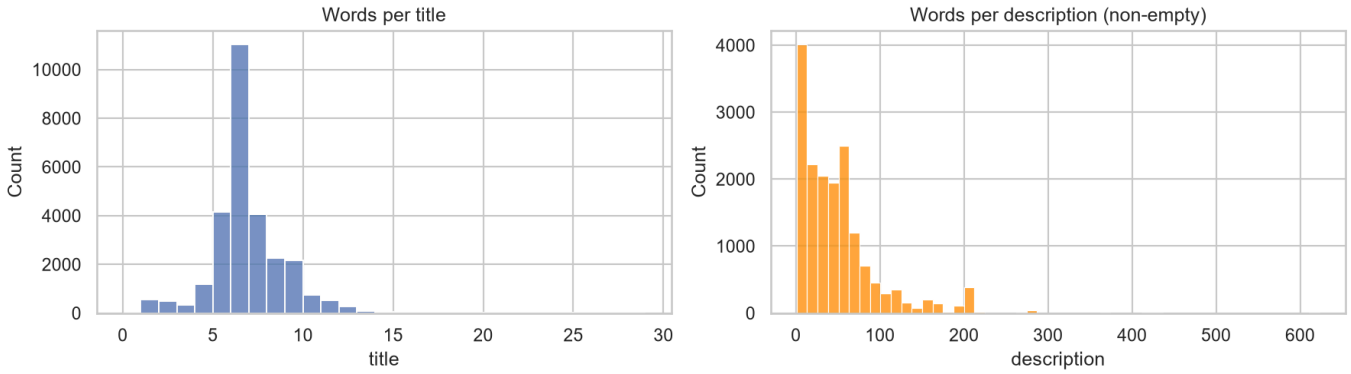


Figure 1. Number of words per title and per non-empty description.

2.3 Most frequent words

The most frequent terms after pre-processing are *neck* (41,145 occurrences), *cotton* (33,252), *tshirt* (32,008), *round*, *women*, *men*, *sleev*, *solid*, *print* and *regular*: together they describe the typical product, a printed cotton round-neck t-shirt.

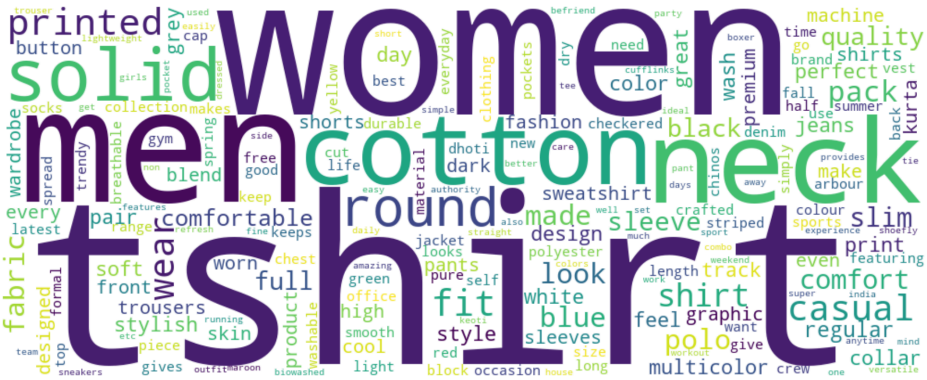


Figure 2. Word cloud of titles and descriptions (after stop-word removal, without stemming).

These terms appear in a large part of the corpus and will get a low IDF, rarer terms such as *sweatshirt* or *jean* will weigh more in the ranking.

2.4 Price, discount and rating

The price is plotted on a log scale because a few expensive products stretch the axis: the median is 545 while the mean is 705.64 (maximum 7,999). Discounts are the norm (median 52%, up to 87%) and ratings are concentrated around the median of 3.8.

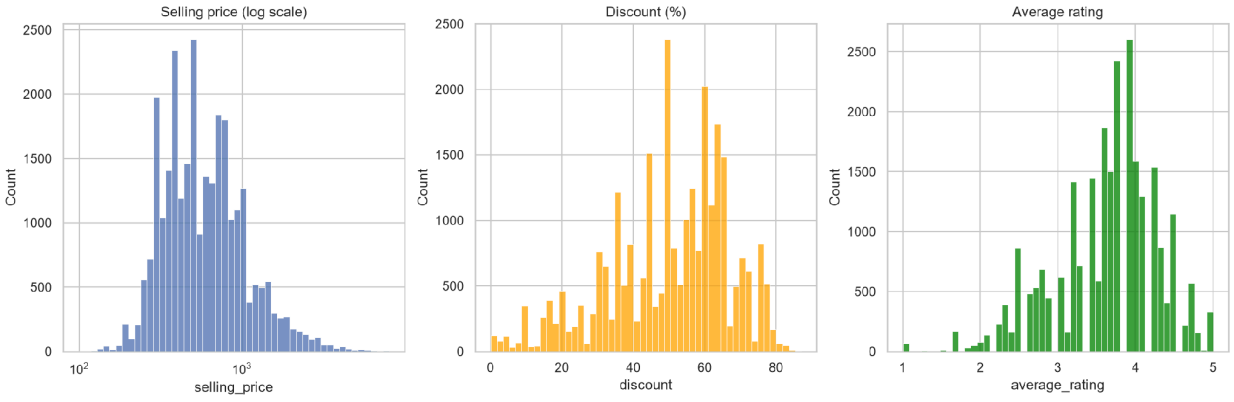


Figure 3. Distribution of selling price (log scale), discount and average rating.

Ranking the products by each field shows that they measure different things. The top-rated products (5.0) are cheap items with large discounts. The most expensive ones (7,800-8,000: a track suit, a blazer, a jacket) have small discounts and ratings from 2.6 to 4.5. The largest discounts (86-87%) are cheap printed t-shirts of a single brand.

2.5 Brands, sellers and stock

The most frequent brands are *ECKO Unl* (951), *Free Authori* (860), *ARBO* (806), *REEB* (802) and *Pu* (798). The top sellers are *RetailNet* (1,416), *SandSMarketing* (887) and *BioworldMerchandising* (842). No brand or seller covers more than 5.4% of the corpus.

	Products	Mean price	Mean discount	Mean rating
In stock	26,436 (94.1%)	686.73	50.2%	3.62
Out of stock	1,644 (5.9%)	1,009.93	48.6%	3.92

2.6 Data quality

- **22,127 repeated titles**: products with the same title differ in colour, size or seller.
- **13,039 products** have Women in the title while their URL says men. Gender is inconsistent across fields, so no single field can be trusted for it.
- **1,643 of the 1,644 out-of-stock products have no seller**, so a missing seller is linked to stock rather than random.

AI usage statement

We used Claude (Anthropic) as an assistant to get initial knowledge of the structure of the whole project before doing the part 1, also for brainstorming, both for EDA analysis and data preparation. Moreover, it helped writing the README.md and reviewed the code and results of the notebook. All decisions and results were checked by the team.